Practical 1: Getting Started with Tableau and Building Basic Visualizations

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a) Connect to a CSV dataset and explore the Tableau interface
b) Identify and differentiate Measures and Dimensions
c) Convert fields between discrete and continuous and observe changes
d) Create and customize a basic bar chart
e) Build a line chart for time-series analysis
f) Create a geographic map visualization
g) Combine multiple charts into a simple interactive dashboard

Dataset: Superstore .

kaggle link: https://www.kaggle.com/code/rajatraj0502/superstore-dataset

Part A: Connect to Dataset & Explore the Interface

1. Launch Tableau Desktop on your machine .
2. On the left-hand blue navigation pane under Connect, click on Text File (Tableau treats CSVs as text files).
3. Select your CSV file from your Mac Finder and click Open.
4. You will be taken to the Data Source page. Here, preview your data rows at the bottom to ensure they look correct, then click Sheet 1 at the bottom left to enter your workspace. Tableau will bypass its faulty drivers, create a native workspace called Clipboard, and instantly open up **Sheet 1**.

Navigating the Interface

- **The Data Pane (Far Left):** This holds all your variables from the dataset.
- **Shelves (Top/Center):** The **Columns** and **Rows** shelves act as your X and Y coordinates.
- **The Marks Card (Left of center):** This controls your visual design properties like color, text labels, and sizes.

Part B: Identify and Differentiate Measures and Dimensions

Look closely at your **Data Pane** on the left. Tableau reads the metadata of the Superstore CSV and splits your items automatically:

- **Dimensions (Categorical / Blue Icons):** Fields containing text, dates, or geography. In Superstore, look for fields like Category, Segment, State, or Order Date. They slice your data into distinct groups.
- **Measures (Numerical / Green Icons):** Fields containing raw quantitative values. Look for fields like Sales, Profit, or Quantity. Tableau will automatically summarize or average these values whenever you place them on a chart.

Part C: Convert Fields Between Discrete and Continuous

Tableau colors its data pills to represent how it visualizes information. **Blue means Discrete** (separate header blocks) and **Green means Continuous** (an infinite unbroken axis).

1. Find the Quantity field in your green measures list.
2. Right-click (or hold Ctrl and click) on Quantity and choose **Convert to Discrete**. Notice its icon changes from a green # to a blue #.
3. Drag this new blue Quantity pill onto your **Columns** shelf. It will draw isolated individual header columns for each exact number (1, 2, 3, etc.).
4. Drag it back off the shelf. Right-click Quantity again and select **Convert to Continuous**.
5. Drag it back onto **Columns**. Instead of individual column blocks, Tableau now draws a singular, smooth, continuous numeric axis line.

Part D: Create and Customize a Basic Bar Chart

1. From your Data Pane, find the Category dimension and drag it up onto the **Columns** shelf.
2. Find the Sales measure and drag it up onto the **Rows** shelf. You will instantly get three vertical bars representing Furniture, Office Supplies, and Technology.
3. **Color customization:** Take Category from the Data pane a second time, and drop it directly onto the **Color** tile in the Marks card.
4. **Label customization:** Drag the Sales pill from the data pane and drop it onto the **Label** tile in the Marks card to display the precise revenue numbers over the bars.
5. Double-click the bottom tab sheet name Sheet 1 and rename it to **Category Bar Chart**.

Part E: Build a Line Chart for Time-Series Analysis

1. Click the **New Worksheet** button at the very bottom tab bar (the small icon with a chart and a plus symbol).
2. Drag the Order Date dimension onto your **Columns** shelf.
3. Drag the Sales measure onto your **Rows** shelf. Because a date field is on your columns shelf, Tableau defaults straight into a line chart.
4. **Drilling Down:** Look at the blue YEAR(Order Date) pill sitting in your Columns shelf. Click the small + **icon** located on that blue pill. It will expand your data into Quarters and Months, revealing hidden sales trends over time.
5. Double-click the sheet tab at the bottom and rename this worksheet to **Sales Trend**.

Part F: Create a Geographic Map Visualization

1. Click the **New Worksheet** button at the bottom to create your third sheet.
2. Look through your Data Pane and find the State field. It should display a tiny globe icon next to it, indicating geographic data.
3. **Double-click** the State field. Tableau will automatically inject Latitude and Longitude coordinates into your shelves and draw a map.
4. **Styling the Map:** Go to your Data Pane, pick up Profit, and drop it directly onto the **Color** icon inside your Marks card. This will instantly color-code each state based on its performance.
5. Rename this worksheet tab at the bottom to **State Profit Map**.

Part G: Combine Charts into an Interactive Dashboard

1. Look at the bottom navigation tab bar and click the **New Dashboard** icon (it looks like a small 2x2 grid window pane with a plus symbol).
2. On the left side panel, you will see a list of your worksheets: *Category Bar Chart*, *Sales Trend*, and *State Profit Map*.
3. Drag your **Category Bar Chart** sheet onto the empty dashboard canvas area.
4. Drag your **Sales Trend** sheet and drop it right along the bottom edge of the bar chart workspace.
5. Drag your **State Profit Map** and drop it alongside the right side of the layout.

Turning on the Magic: Interactivity

1. Click once inside your **Category Bar Chart** zone on the dashboard to select it.
2. Look at the subtle gray menu options border that opens up on its top-right corner.
3. Click the tiny **Funnel Icon** (**Use as Filter**).

Now test out your creation! Click on the **Technology** bar in your bar chart. The map and the time-series line chart will dynamically isolate and update to display data *exclusively* for your Technology sales. Click the bar a second time to reset the filter.